



Analog IC Design

CMOS Technology

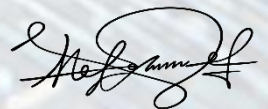
MOHAMMED NASR EL-DIN

Project 02

Fully Differential Folded OTA
Supervised by Dr. Hesham Omran



cādence[®]

A handwritten signature in black ink, likely belonging to Dr. Hesham Omran, is located in the bottom right corner of the cover.

Summary

This report presents Project 02 of the Analog IC Design (CMOS Technology) Summer Training program at the Information Technology Institute (ITI), supervised by Dr. Hesham Omran.

The objective of this project is to design, size, and simulate a Fully Differential Folded Cascode OTA using the g_m/I_D design methodology in a $180nm$ CMOS technology ($GF180MCU$), meeting a target specification set including DC loop gain ($\geq 60dB$), phase margin ($\geq 70^\circ$), common-mode input range, differential output swing ($1.2V_{pk-pk}$), and closed-loop settling time ($100ns$ at 1% error) constraints at a $2.5V$ supply.

The work covers the full design flow: deriving the required open-loop transconductance and gain-bandwidth product from the closed-loop gain ($A_{cl} = 2$), the feedback factor ($\beta = 1/3$), and the settling-time requirement; extracting design points for all transistors using g_m/I_D charts; and sizing the PMOS input differential pair, the common-source and common-gate branch current sources, the tail current sources, the NMOS and PMOS cascode transistors, and the current mirrors — followed by the design of both a behavioral and an actual common-mode feedback (CMFB) network.

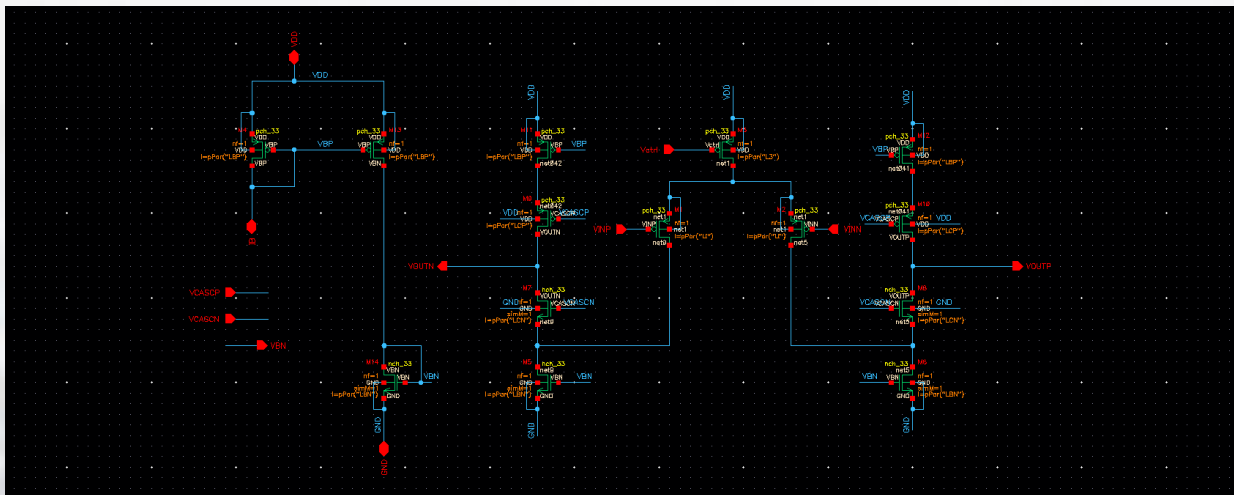
The design was verified through open-loop and closed-loop simulations covering the DC operating point, differential AC response, loop-gain (STB) analysis for both the differential and common-mode feedback loops, and transient analysis (differential and common-mode input pulses, output swing, and 1% settling time), using Cadence Virtuoso for schematic capture and simulation and ADT (Analog Designer's Toolbox) for g_m/I_D based sizing. Simulation results closely matched hand calculations for DC gain, bandwidth, and GBW; the final design achieved smaller than 1% settling time of $96.45ns$ and a differential output swing close to the $1.2V_{pk-pk}$ target, though the common-mode feedback loop's phase margin required additional tuning of the CMFB input transistor sizing to reach an adequate stability margin.

We will use a unity current split $S = I_{B1}/I_{B2} = 1$ between the input and folding branches to maintain a balanced distribution of bias current, providing high slew rate, symmetric slewing behavior, and good phase margin. Although a higher split ratio may offer better bandwidth, gain, and noise performance, $S = 1$ provides a balanced overall behavior and avoids starving the folding branch. For further details on the effect of the current split ratio and its optimization, refer to the paper. [H. Omran, "Optimum Split Ratio for Folded Cascode OTA Bias Current: A Qualitative and Quantitative Study," Des 2019.](#) (it's preferred to open the link using phone device if it didn't run on computer device)

Specs and Circuit Topology

Technology	0.13um	GF180MCU
Supply voltage	1.2V	2.5V
Closed loop gain	2	2
Phase margin at the required ACL	$\geq 70^\circ$	$\geq 70^\circ$
CM input range – low	≤ 0	≤ 0
CM input range – high	$\geq 0.6V$	$\geq 1V$
Differential output swing	0.6Vpk-to-pk	1.2Vpk-to-pk
Load	500fF	500fF
DC Loop gain	50dB	60dB
CL settling time for 1% error	100ns	100ns

Since the input common mode range is closer to GND rail, the better choice for the input pair is PMOS device.



Process **TT** at **nominal** temperature.

The technology used: **GF180nm**

DC voltage supply = **2.5V**, Two DC sources for cascode transistors, and an DC source for CM output level.

Used an external **10μA** DC reference current source.

M1, M2: CS branch input pair

M3: CS branch current source

M4: Mirror current source

M5, M6: tail current source

M7, M8: NMOS cascode transistors

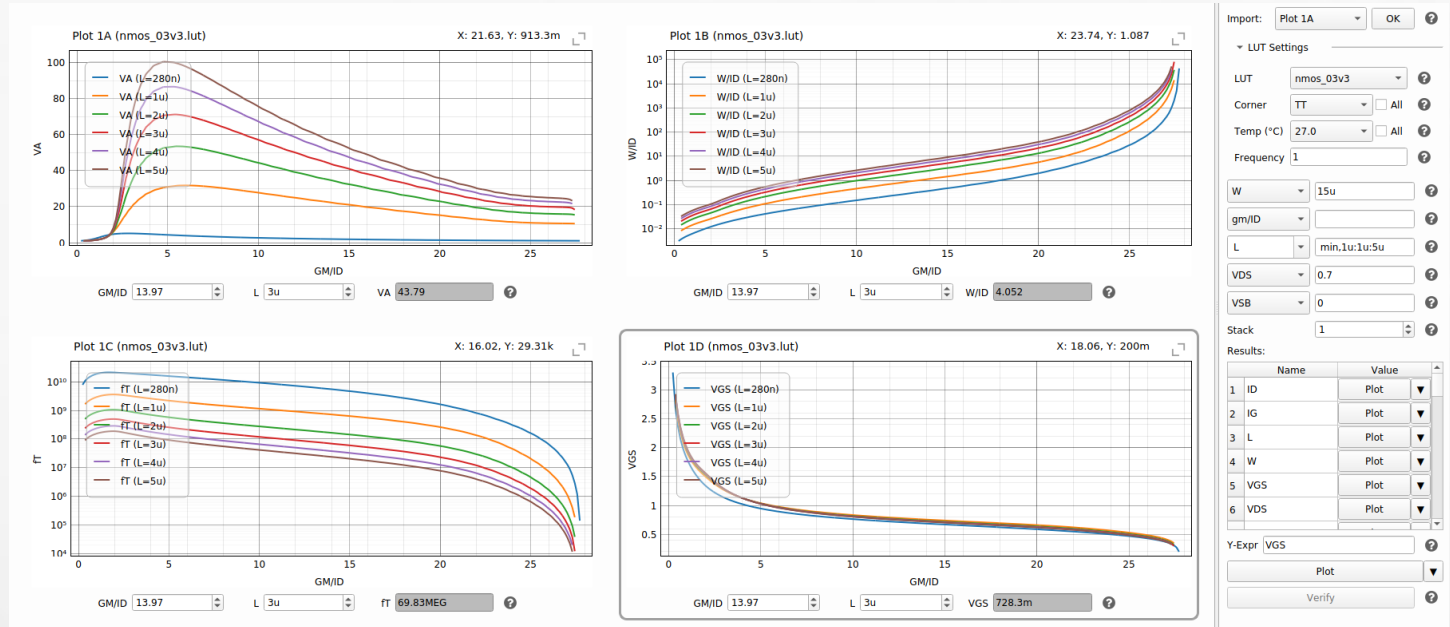
M9, M10: PMOS cascode transistors

M11, M12: CG branch current source

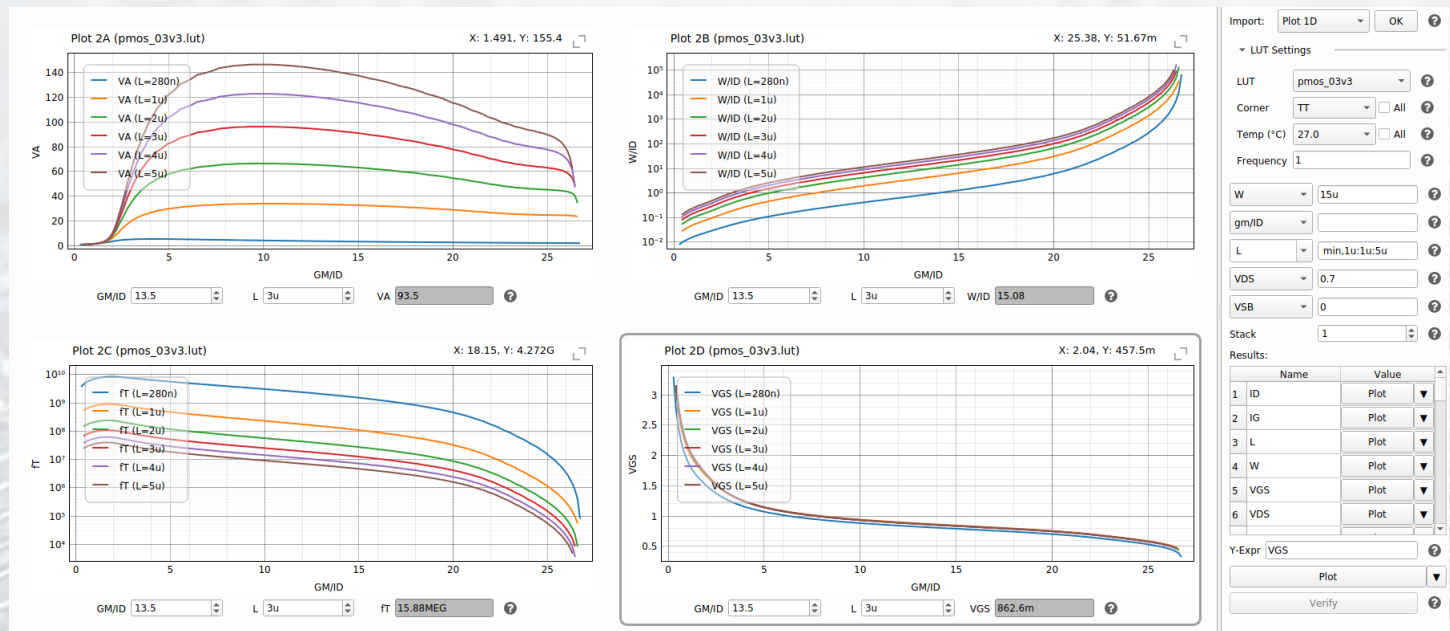
M13, M14: PMOS to NMOS current mirror

g_m/I_D Design Charts

Since all of the following figure of merits (FOMs) " V_A , $\frac{W}{I_D}$, f_T and V_{GS} " almost doesn't depend on the MOSFET width, assume constant width to plot them $W = 15\mu m$, let $V_{DS} = 0.7, V_{SB} = 0$



NMOS charts



PMOS charts

OTA Design

The closed loop gain is given by $A_{cl} = \frac{Z_F}{Z_S} \approx \frac{C_S}{C_F} = 2$, let $C_S = 2pF \Rightarrow C_F = 1pF$

For the inverting amplifier topology, $\beta = \frac{Z_S}{Z_S + Z_F} \approx \frac{1}{1 + C_S/C_F} = \frac{1}{3}$

For 1% closed loop settling time error, let $\frac{\omega_{p2}}{\omega_u} \rightarrow \infty \Rightarrow \frac{t_s}{\tau_{cl}} = 4.6 \Rightarrow \tau_{cl} = \frac{100ns}{4.6} \approx 21.74ns$, $BW_{cl} = \frac{1}{2\pi \tau_{cl}} \approx 7.32MHz$

Open loop GBW = $\frac{1}{\beta} \times BW_{cl} \approx 21.96MHz$, closed loop GBW = $A_{cl} \times BW_{cl} \approx 14.64MHz$

Open loop GBW = $\frac{g_{m1,2}}{2\pi C_{out}} = 21.96MHz$, $C_{out} = C_L + C_{OTA_{out}} + \left[\frac{C_F \times (C_S + C_{OTA_{in}})}{C_F + (C_S + C_{OTA_{in}})} \right] + C_P$

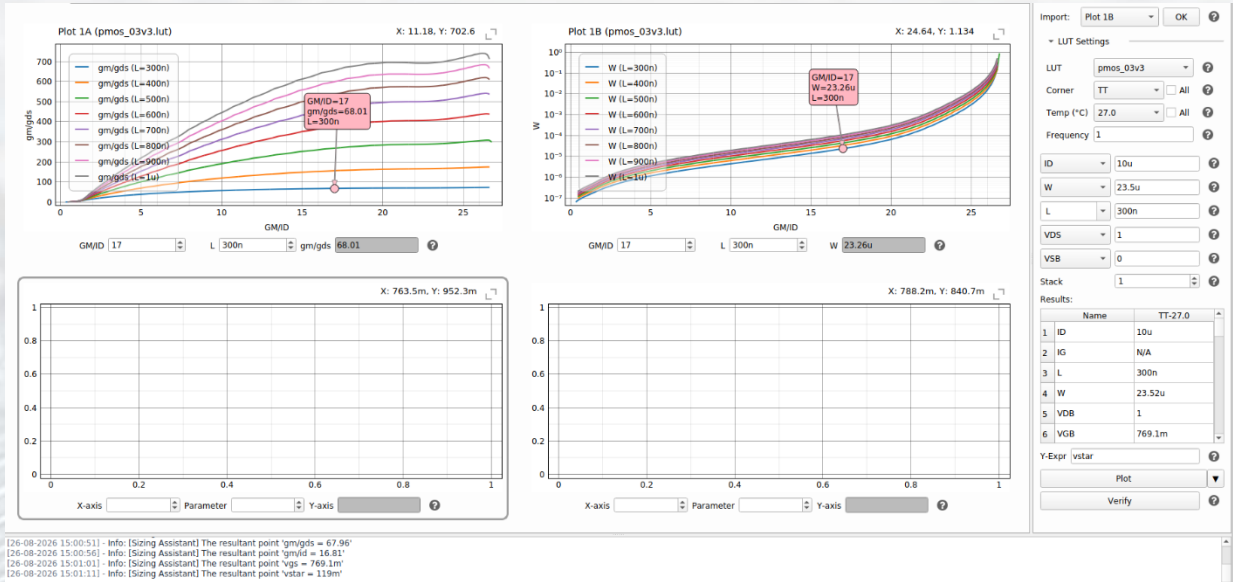
Neglecting the parasitic capacitance C_P and the OTA capacitances $C_{OTA_{out}}$ & $C_{OTA_{in}} \Rightarrow C_{out} \approx C_L + \left[\frac{C_F \times C_S}{C_F + C_S} \right] \approx 1.167pF$

$\Rightarrow g_{m1,2} = 21.96MHz \times 2\pi C_{out} \approx 161.02\mu S$, take $g_{m1,2} = 170\mu S$

We will directly assume values for L and g_m/I_D (or V^*) based on your designer's intuition and folded cascode trade-offs matrix. These assumptions will greatly simplify the design process and can be later tuned in simulations

For the PMOS input pair $M_{1,2}$, assume small $L = 300nm$ and bias it in MI or WI $\frac{g_m}{I_D} = 17\mu S/\mu A$. This maximizes the GBW and minimizes the input capacitive loading $\Rightarrow I_{D1,2} = 10\mu A$

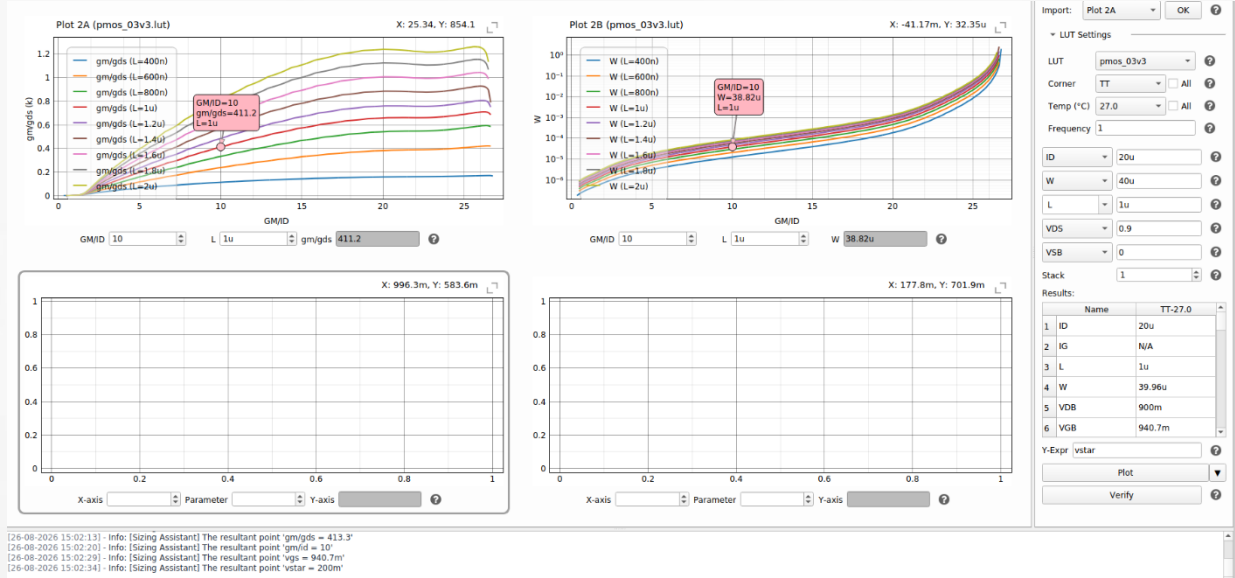
$V_{SB1,2} = 0$, let $V_{DS1,2} = 1V$



PMOS input pair design point: $L = 300nm$, $W = 23.5\mu m$, $V^* = 120mV$

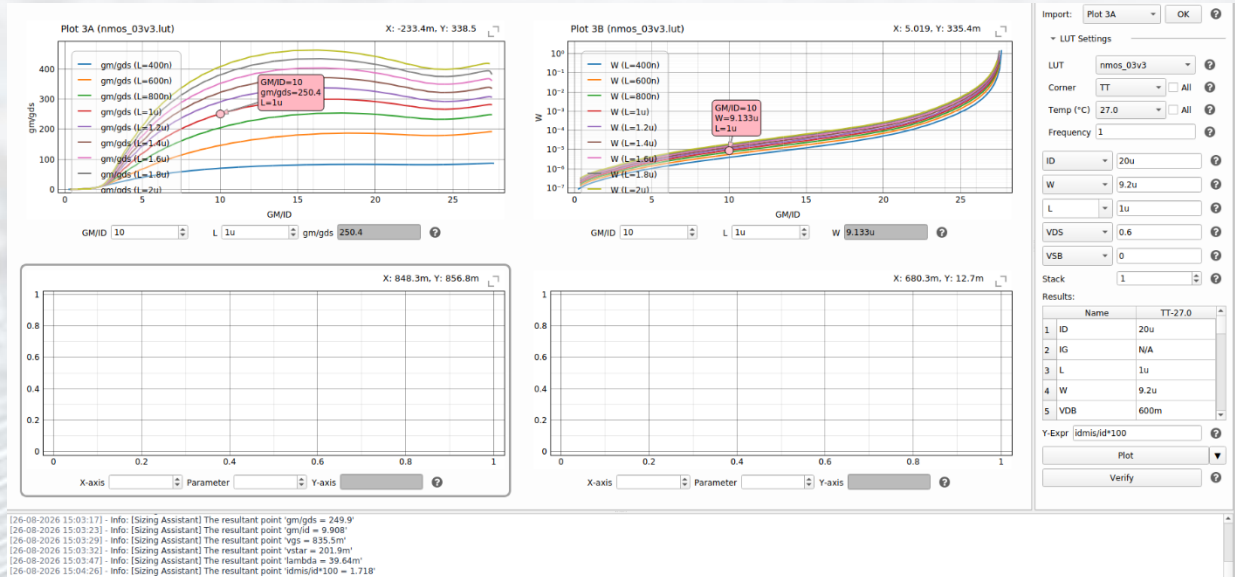
For current sources, use relatively long $L = 1\mu m$ and bias them in SI $\frac{g_m}{I_D} = 10\mu S/\mu A$, a large g_m will not help the gain but will increase the offset and noise.

For the NMOS CS branch current source M_3 , $I_D = 20\mu A$, $L = 1\mu m$ and $\frac{g_m}{I_D} = 10\mu S/\mu A$, $V_{SB} = 0$, let $V_{DS} = 0.9V$



PMOS CS branch current source design point: $L = 1\mu m$, $W = 40\mu m$, $V^* = 200mV$

For NMOS tail current sources and current mirror transistor $M_{5,6}$ and M_{14} , $I_D = 20\mu A$, $L = 1\mu m$ and $\frac{g_m}{I_D} = 10\mu S/\mu A$, $V_{SB} = 0$, let $V_{DS} = 0.6V$

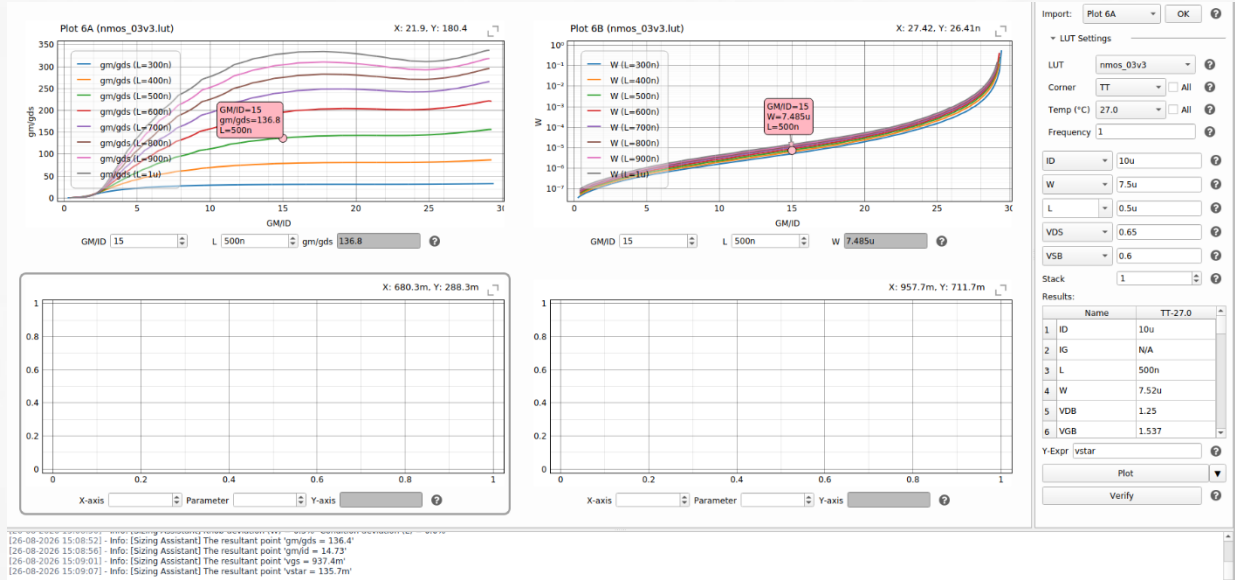


PMOS CS branch current source and current mirror transistor design point: $L = 1\mu m$, $W = 9.2\mu m$, $V^* = 202mV$

The results gives less than 2% mismatch "1.718%", and less than 4% current change for each 1V $\lambda = 36.64mV < 40mV$

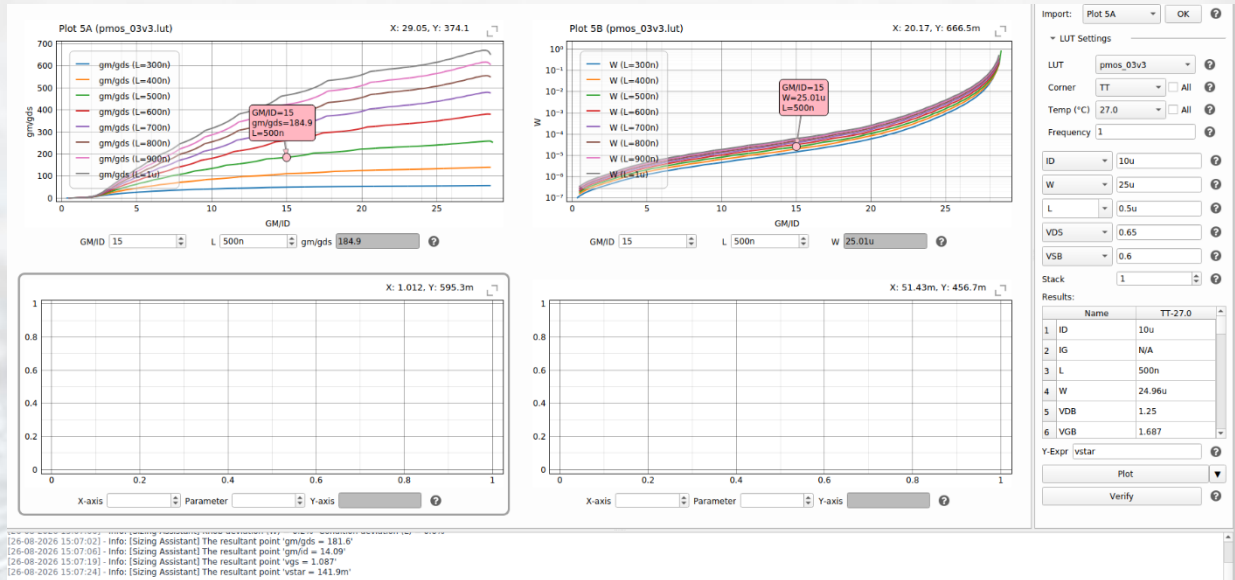
For cascode transistors, use moderate $L = 0.5\mu\text{m}$ and bias them in MI or WI $g_m/I_D = 15$, a large g_m helps the gain and doesn't increase the noise.

For NMOS cascode transistors $M_{7,8}$, $I_D = 10\mu\text{A}$, $L = 0.5\mu\text{m}$ and $\frac{g_m}{I_D} = 15\mu\text{S}/\mu\text{A}$, let $V_{DS} = 0.65\text{V}$, $V_{SB} = 0.6$



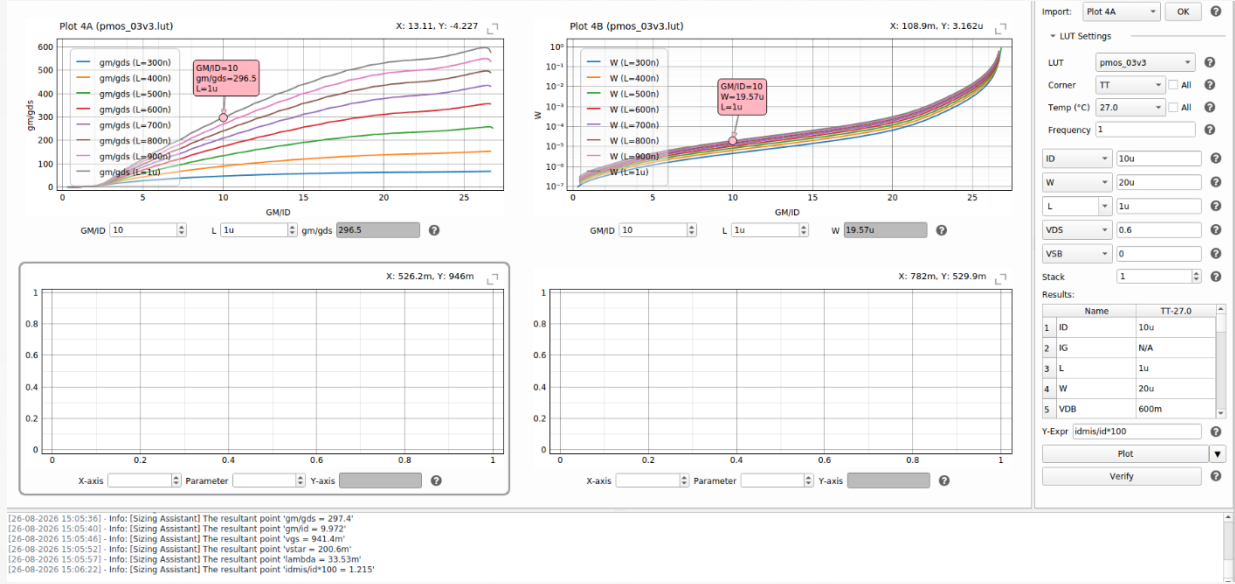
NMOS cascode transistor design point: $L = 0.5\mu\text{m}$, $W = 7.5\mu\text{m}$, $V^* = 135.7\text{mV}$

For PMOS cascode transistors $M_{9,10}$, $I_D = 10\mu\text{A}$, $L = 0.5\mu\text{m}$ and $\frac{g_m}{I_D} = 15\mu\text{S}/\mu\text{A}$, let $V_{DS} = 0.65\text{V}$, $V_{SB} = 0.6$



PMOS cascode transistor design point: $L = 0.5\mu\text{m}$, $W = 25\mu\text{m}$, $V^* = 142\text{mV}$

For PMOS tail current sources and current mirror transistor $M_{11,12}$ and M_4 , $I_D = 10\mu A$, $L = 1\mu m$ and $\frac{g_m}{I_D} = 10\mu S/\mu A$, let $V_{DS} = 0.65V$, $V_{SB} = 0.6$

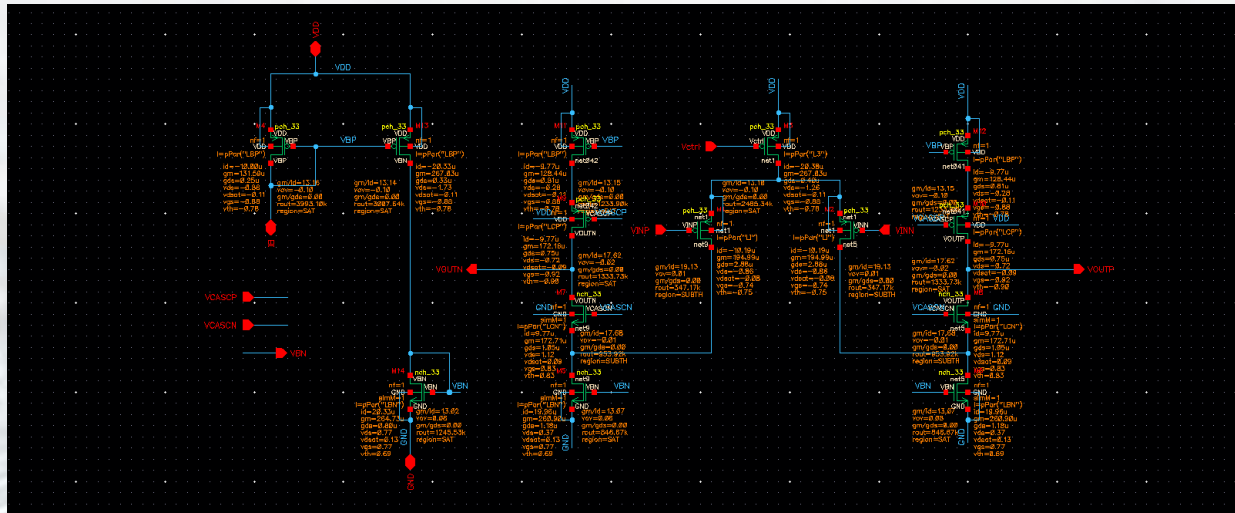


PMOS CG branch current sources and mirror transistor design point: $L = 1\mu m$, $W = 20\mu m$, $V^* = 200.6mV$

The results gives less than 2% mismatch “1.215%”, and less than 4% current change for each 1V $\lambda = 33.53mV < 40V$

For PMOS current mirror M_{13} $I_D = 20\mu A$, will have the same sizing of M_4 with double width $W = 40\mu m$

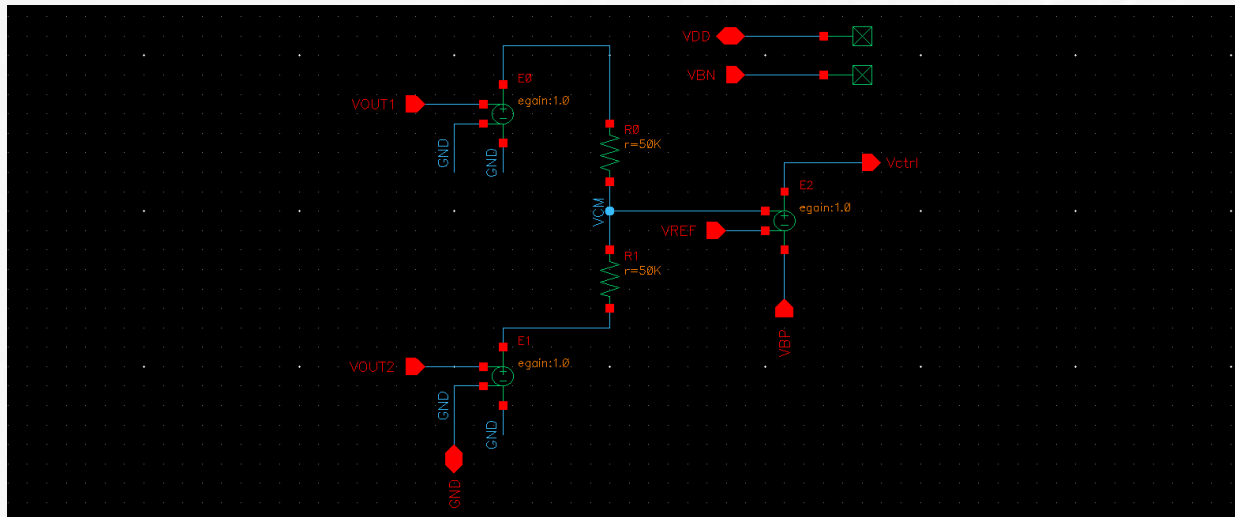
For the DC bias sources $V_{CASCN} \approx V_{GS7,8} + V_{5,6}^* \approx 1.2V$, and $V_{CASCN} \approx V_{DD} - V_{11,12}^* - |V_{GS9,10}| \approx 1.3V$



Pre-DC OP simulation

The above pre-simulation shows over g_m/I_D for all transistors than the designed values, so decreasing the width for all transistor (also the current mirror transistor) with the same ratio (in my case 1/2) will gives smaller $g_m/I_D \propto \sqrt{W}$ for the same current.

Open-Loop OTA Simulation (Behavioral CMFB)

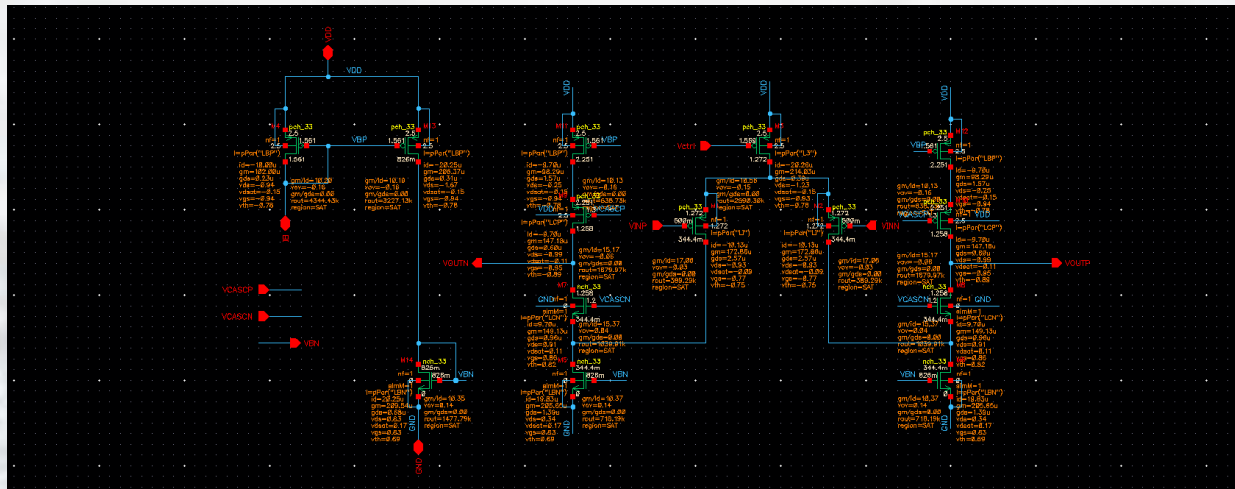


Behavioral CMFB

Set V_{ICM} in the middle of the required CMIR, and V_{REF} to maximize the output swing.

$$\Rightarrow V_{ICM} = 0.5V, V_{REF} = \frac{(V_{DD}-2V^*)+(2V^*)}{2} = 1.25V$$

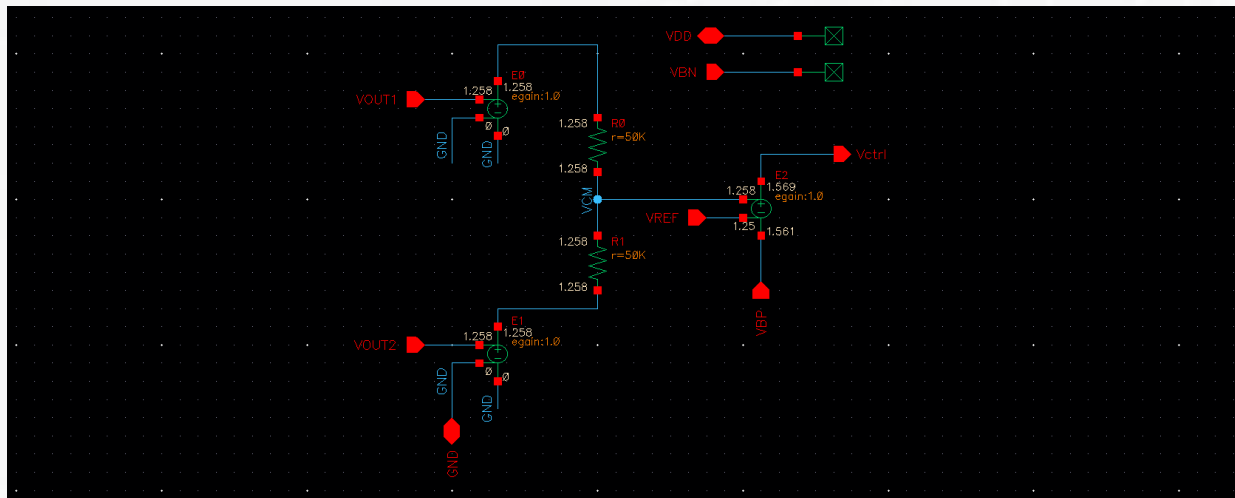
DC OP simulation



Folded OTA DC point annotation

The above results shows that the CM output **1.258V**

The total current consumed by the OTA and the current mirror **59.91μA**



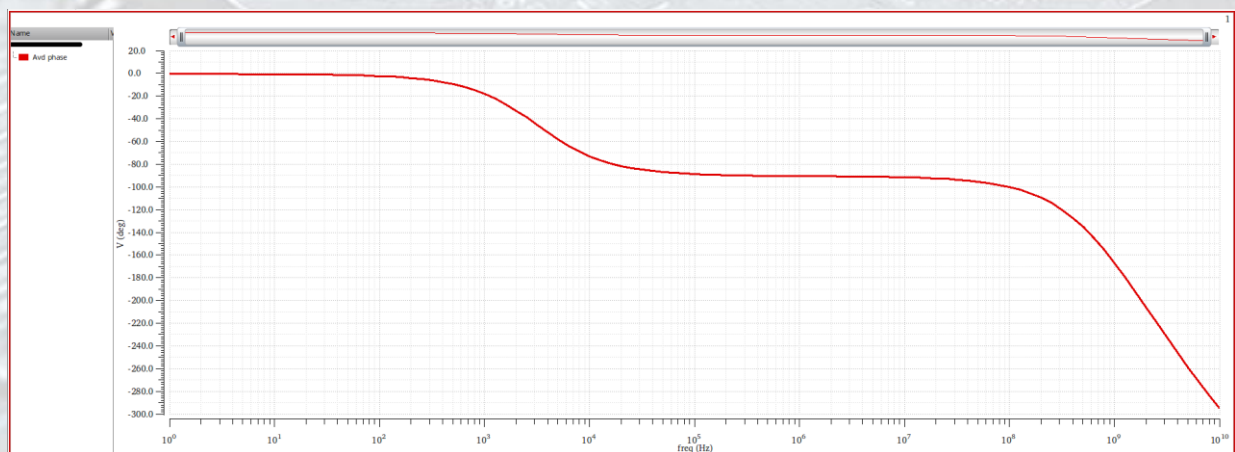
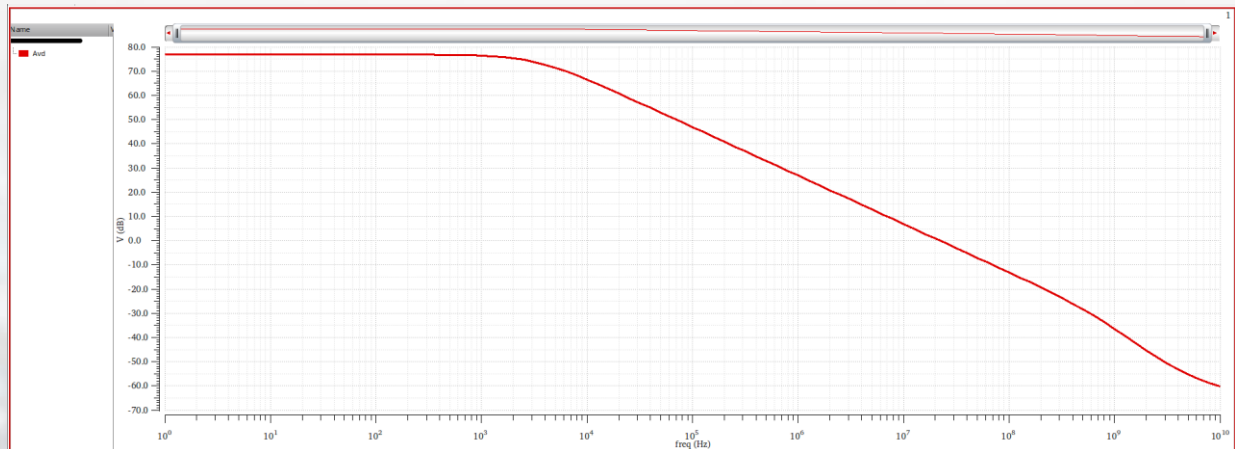
CMFB DC point annotation

For the error amplifier:

The differential input = $1.258 - 1.25 = 8mV$, the differential output = $1.569 - 1.561 = 8mV$

The relation between them is the unity gain of the ideal error amplifier.

Diff small signal



A_{vd}	7.15K
$A_{vd} \text{ dB}$	77.09 dB
GBW	22.63MHz
UGF	22.92MHz
PM	87.84°
BW	3.158KHz

Simulation results

Using $G_m R_{out}$ method:

Let $G_m \approx g_{m1,2} \approx 172.8\mu S$

$$R_{out} \approx r_{o7,8} [g_{m7,8} (r_{o5,6} // r_{o1,2})] // r_{o9,10} (g_{m9,10} r_{o11,12})$$

$$\approx 1041K\Omega [149.13\mu S (720K\Omega // 389.1K\Omega)] // 1680K\Omega (147.18\mu S 637K\Omega) \approx 36.82M\Omega$$

$$\Rightarrow A_{vd} \approx G_m R_{out} \approx 172.8\mu S \times 36.82M\Omega \approx 6.4K \equiv 76.12dB$$

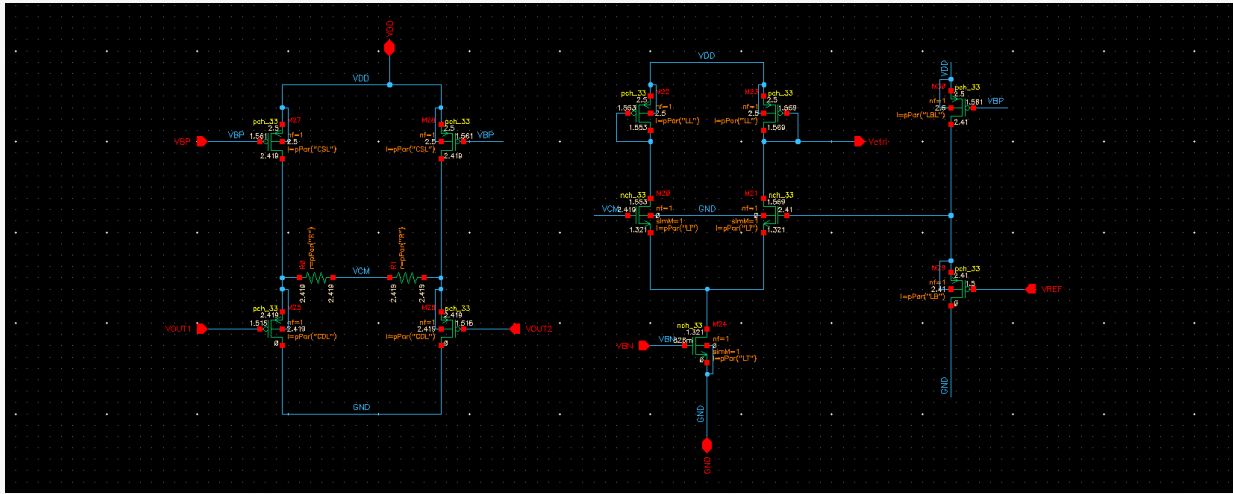
$$GBW \approx \frac{g_{m1,2}}{2\pi C_{out}} = \frac{172.8\mu S}{2\pi 1.167pF} \approx 23.56MHz$$

$$BW = \frac{GBW}{A_{vd}} \approx \frac{23.56MHz}{6.4K} \approx 3.68KHz$$

Note the values of g_m and r_o getting from OP simulation

Parameter	Simulation	Hand calculations
DC diff gain	77.09dB	76.12dB
BW	3.16KHz	3.68KHz
GBW	22.63MHz	23.56MHz

Open-Loop OTA Simulation (Actual CMFB)



Actual CMFB

Let the sizing for PMOS $M_{22,23,27,28,29,30}$ loads, current sources and buffer inputs is same as $M_{11,12,4}$ with smaller width to reduce the current consumed by the CMFB, and also the same relation between the NMOS M_{24} tail current source and $M_{5,6}$, taking $W_{CMFB} = \frac{1}{2}W_{FOLDED}$: $W_{22,23,27,28,29,30} = 5\mu m$ and $W_{24} = 2.5\mu A$

For the OTA input transistors, let $L = 1\mu m$, assume the value of I_D and g_m/I_D to select the width of them and then we can tune it to meet the specs.

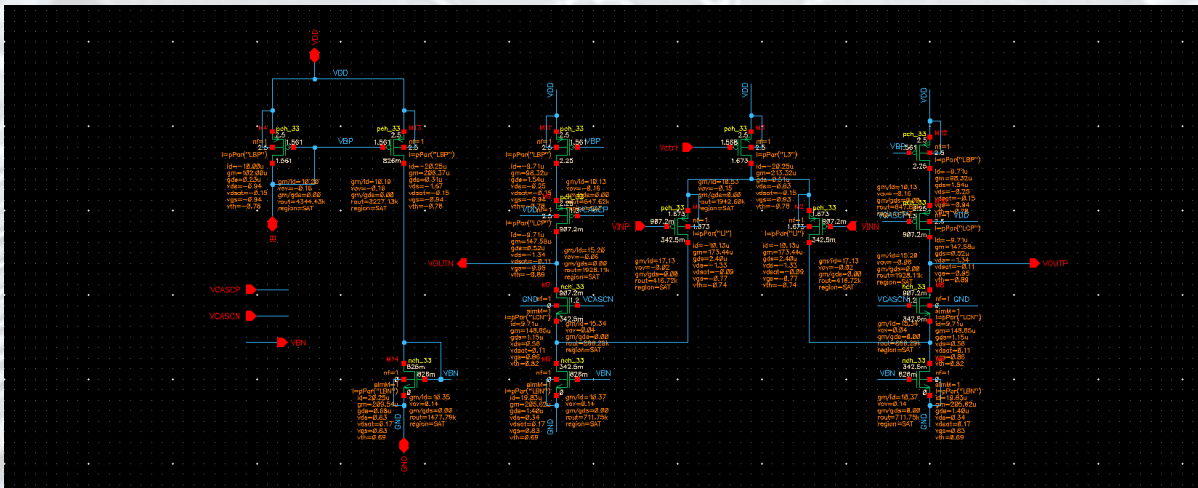
Let $I_D = 5\mu A$, and $g_m/I_D = 15\mu S/\mu A \Rightarrow W = 7.25\mu m$

Set V_{ICM} in the middle of the required CMIR, and V_{REF} to maximize the output swing.

$$\Rightarrow V_{ICM} = 0.5V, V_{REF} = \frac{(V_{DD} - V^* - V_{GS}) + (2V^*)}{2} \approx 0.9V$$

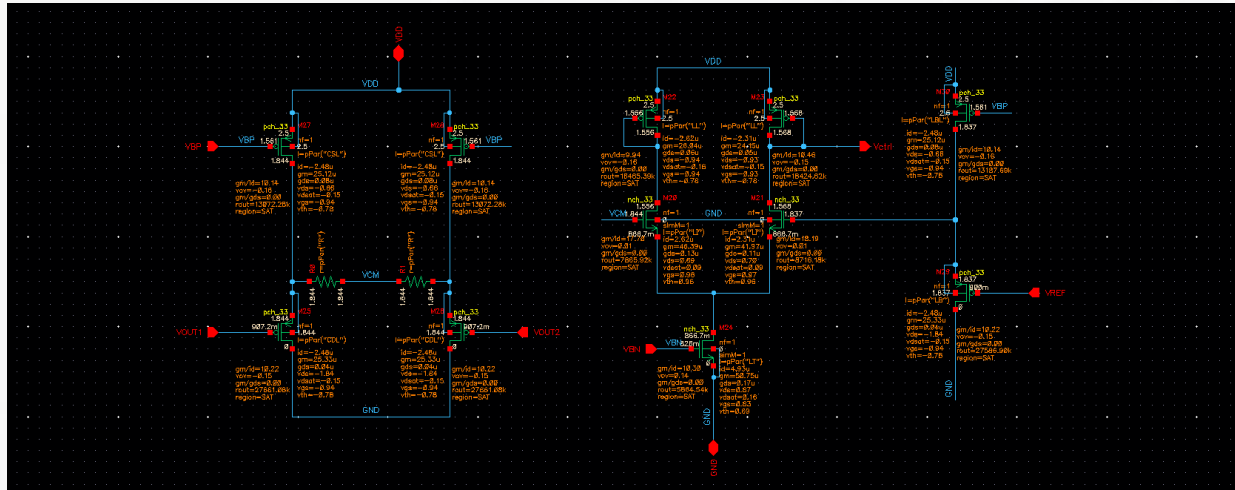
LUT	nmos_03v3	?
Corner	TT	All ?
Temp (°C)	27.0	All ?
Frequency	1	?
ID	5u	?
gm/ID	15	?
L	1u	?
VDS	1	?
VSB	0.8	?
Stack	1	?
Results:		
	Name	TT-27.0
1	ID	5u
2	IG	N/A
3	L	1u
4	W	7.24u

DC OP simulation



Folded OTA DC point annotated

The above results shows that the CM output **907.2mV**, since the CMFB circuit keeps the output CM close to the V_{REF}



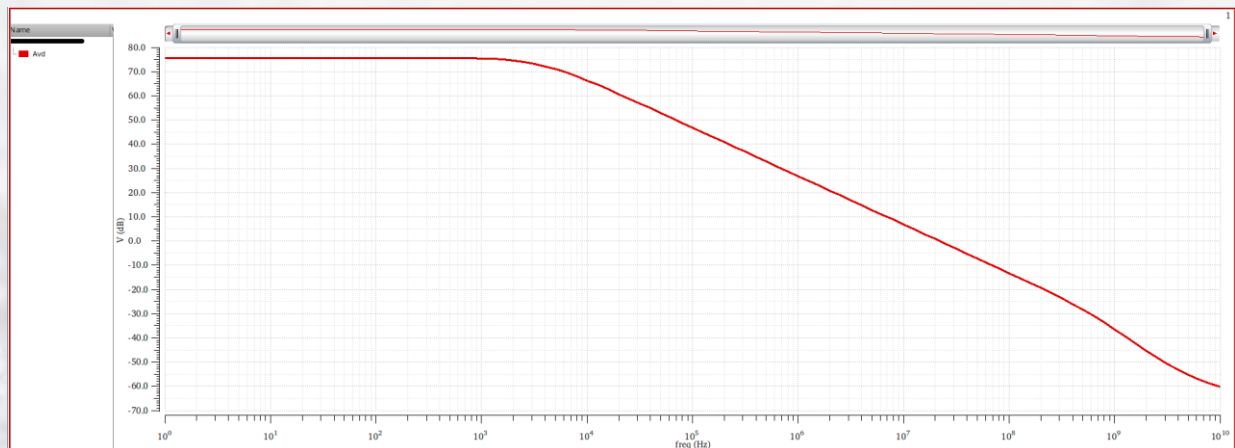
CMFB DC point annotated

The total current consumed in the CMFB is **12.37 μ A**

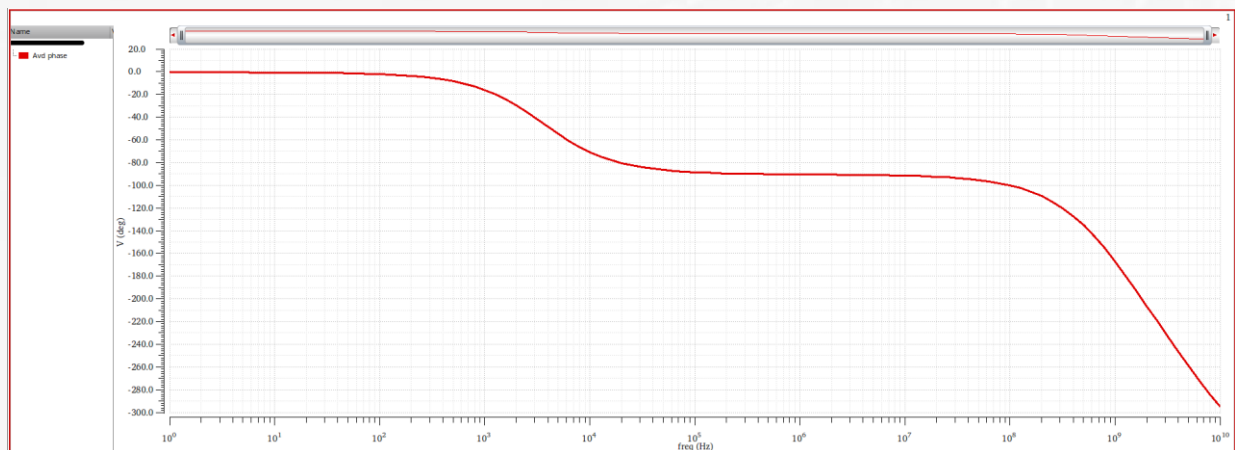
The differential input = $1.844 - 1.837 = 7mV$, the differential output = $1.568 - 1.556 = 12mV$

The relation between them is the gain of the OTA error amplifier, ideally unity.

Diff small signal



A_{vd} dB



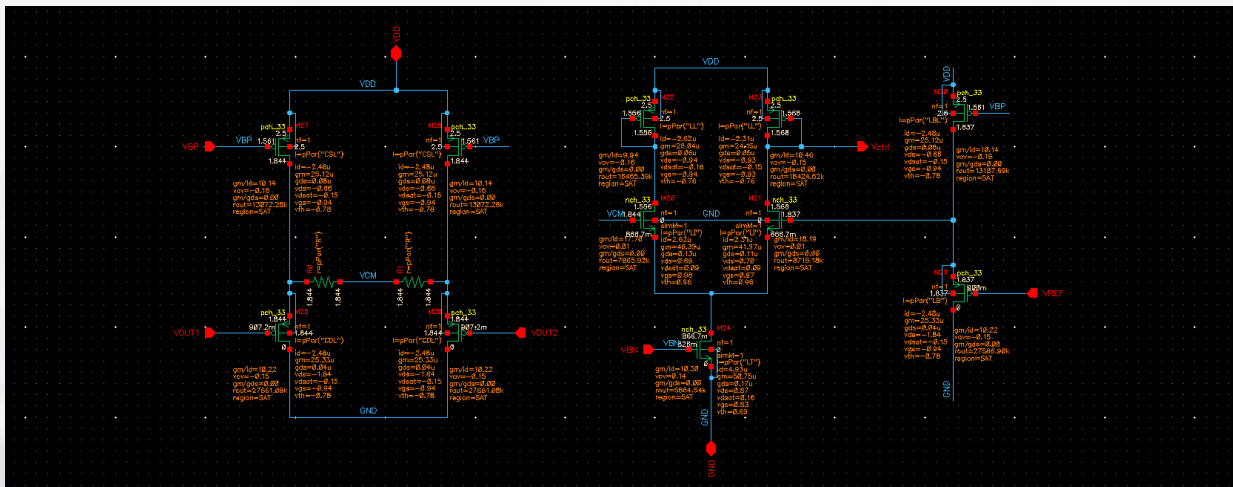
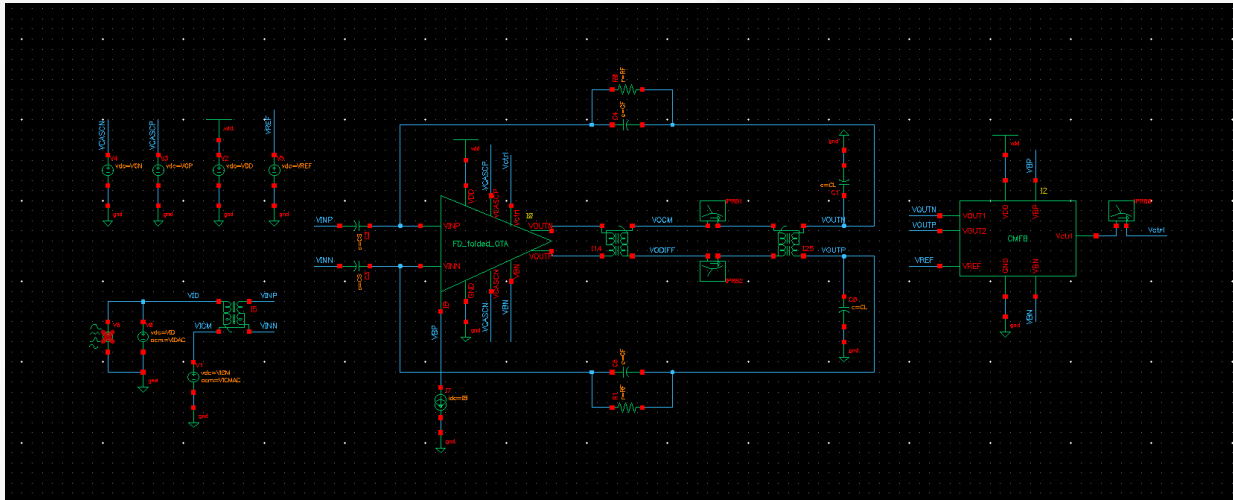
A_{vd} phase

<i>A_{vd}</i>	6.283K
<i>A_{vd} dB</i>	75.96 dB
<i>GBW</i>	22.55MHz
<i>UGF</i>	22.77MHz
<i>PM</i>	87.85°
<i>BW</i>	3.581KHz

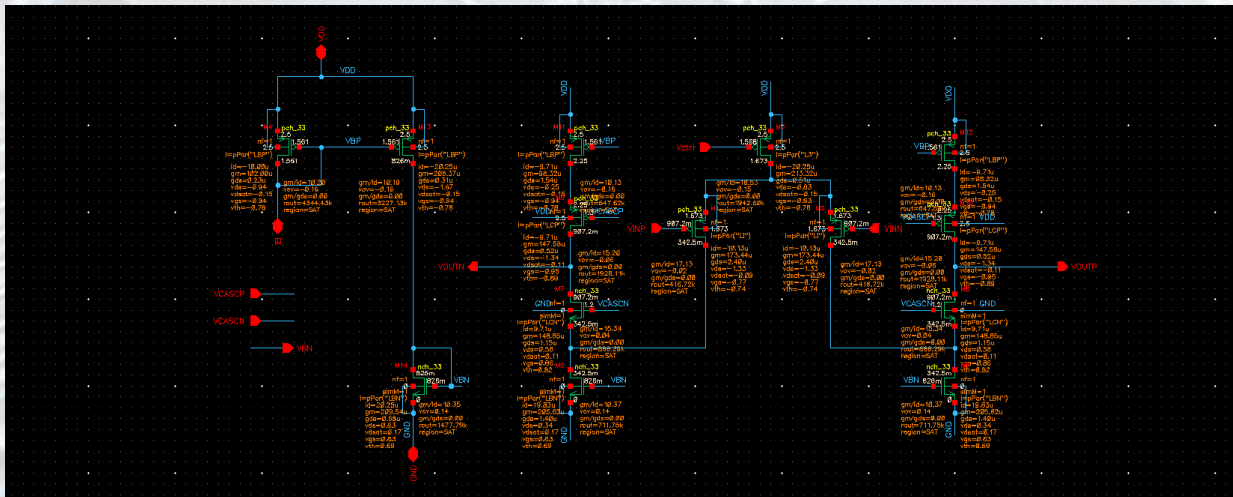
Simulation results

Parameter	Behavioral CMFB	Actual CMFB
DC diff gain	77.09dB	75.96dB
BW	3.16KHz	3.581KHz
GBW	22.63MHz	22.55MHz

DC OP simulation



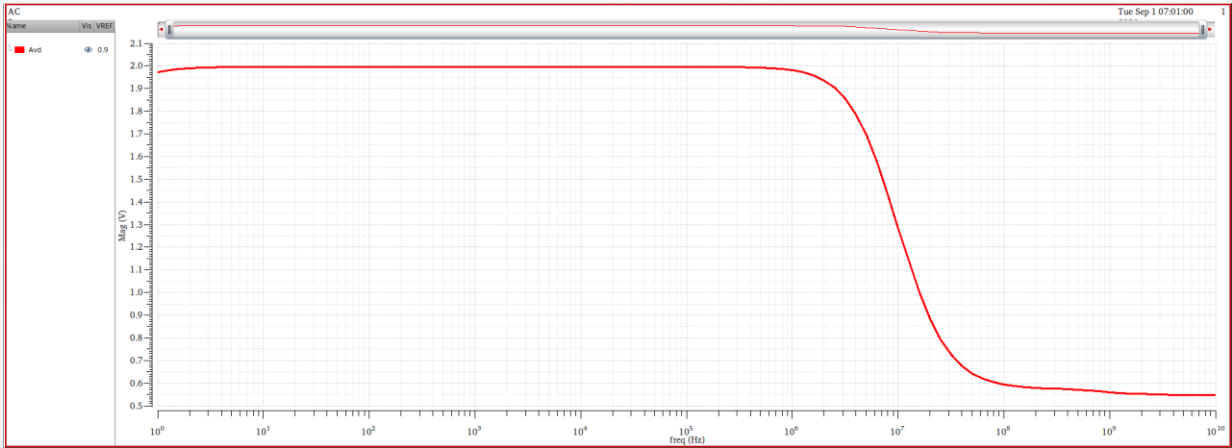
CMFB DC point annotated



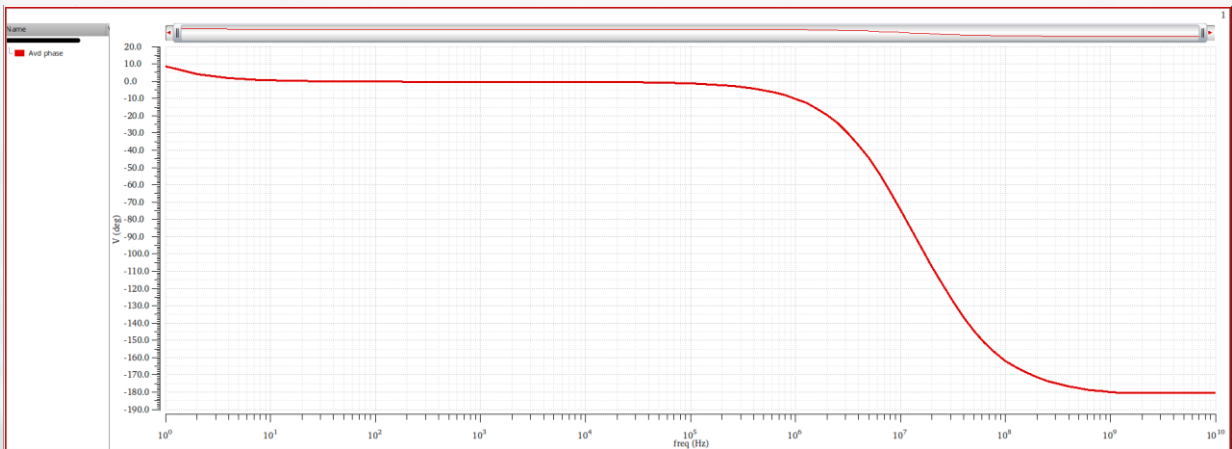
Folded OTA DC point annotated

The above results shows that the CM output is $907.2mV$, since the CMFB circuit keeps the output CM close to the V_{REF} and the CM input is $907.2mV$ same as the output, since C_S act as O.C. $\Rightarrow$ there is no input current, so there will be no current through the feedback network that makes zero voltage drop between the output and the input.

Differential closed-loop response



V_{OD} vs freq



V_{OD} phase

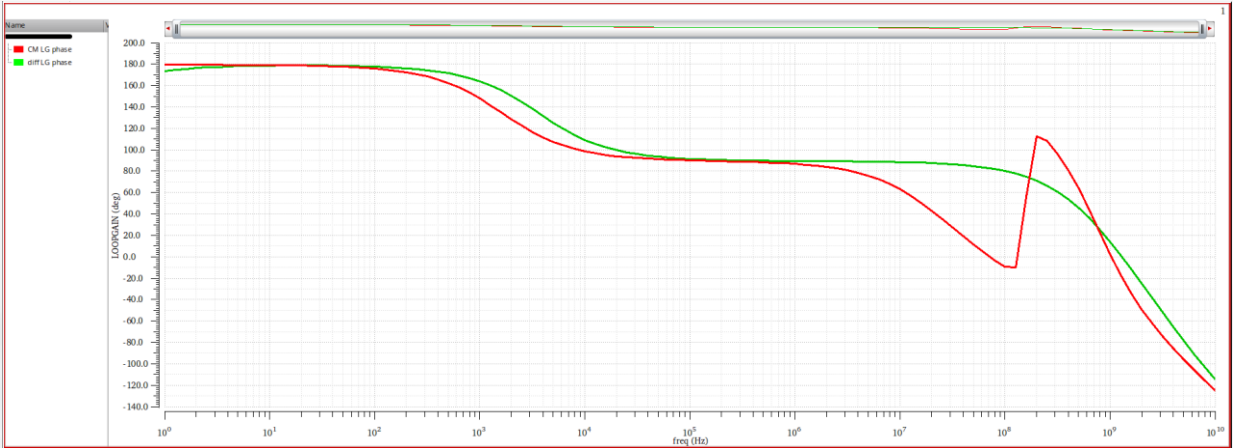
A_{vd}	1.999
A_{vd} dB	6.016 dB
GBW	16.81MHz
UGF	16MHz
PM	83.57°
BW	8.491MHz

Simulation results

Differential and CMFB loops stability



dB gain: ■ diff LG ■ CM LG

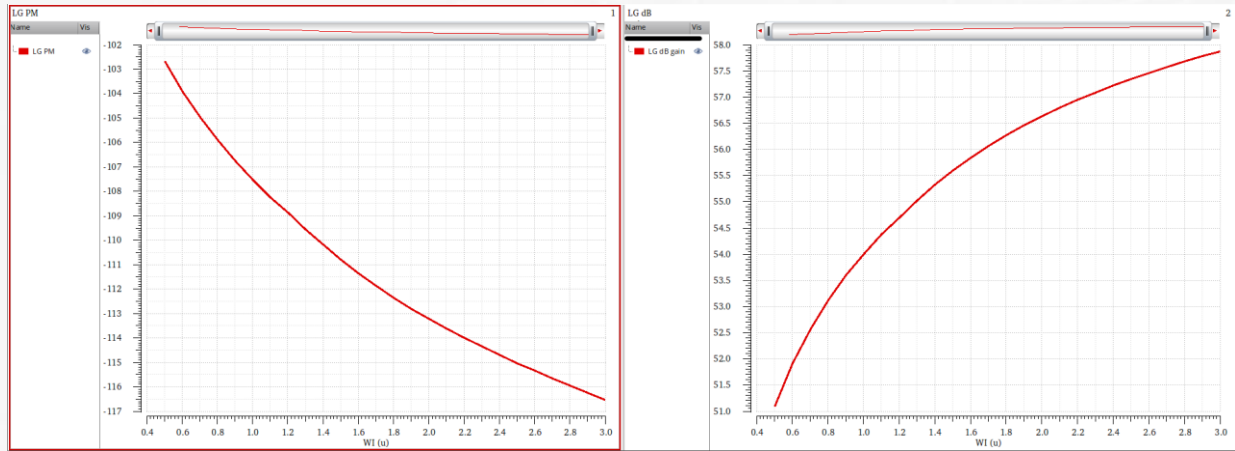


phase: ■ diff LG ■ CM LG

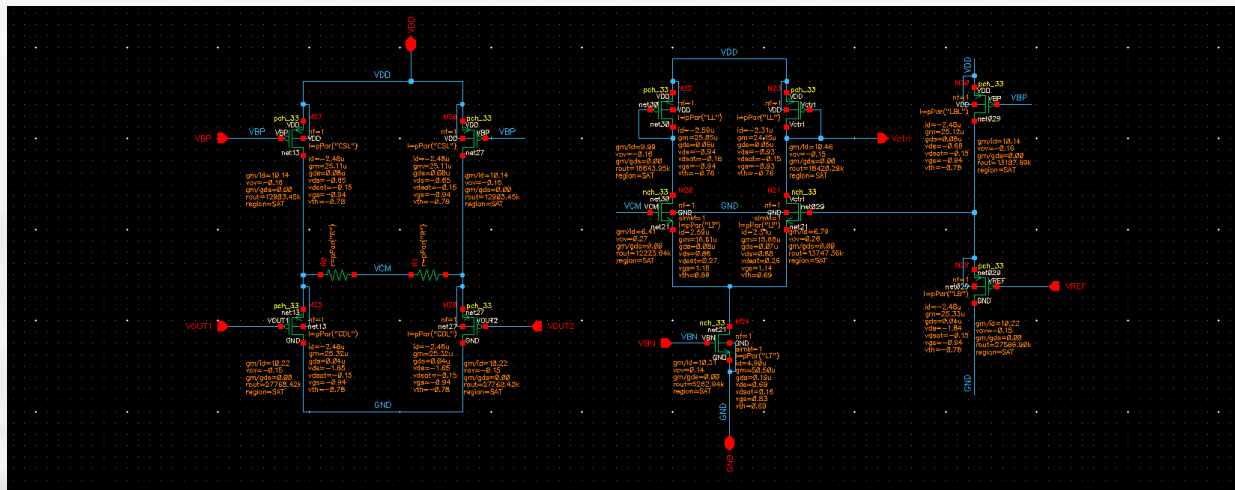
LG	2.093K
LG dB	66.41 dB
GBW	7.365MHz
UGF	7.504MHz
PM	89.33°
BW	3.511KHz
Diff LG Simulation results	
LG	7.833K
LG dB	77.88 dB
GBW	12.69MHz
UGF	11.35MHz
PM	56.4°
BW	1.616KHz

CM LG Simulation results

the $CM\ LG\ PM = 56.4^\circ$ which is less than the desired.



The above graph shows the relation between the CM PM, CMFB dB gain with the width of the CMFB OTA input transistors. Tuning the width of them to be $0.5\mu m$, then check the CMFB DC OP, CL diff small signal and CM LG.



All transistors are in saturation

A_{vd}	1.999
$A_{vd}\ dB$	6.016 dB
GBW	16.81MHz
UGF	16MHz
PM	83.57°
BW	8.492MHz

Closed-loop Diff small signal Simulation results checked

<i>LG</i>	2.88K
<i>LG dB</i>	69.19 dB
<i>GBW</i>	4.657MHz
<i>UGF</i>	4.612MHz
<i>PM</i>	77.5°
<i>BW</i>	1.613KHz

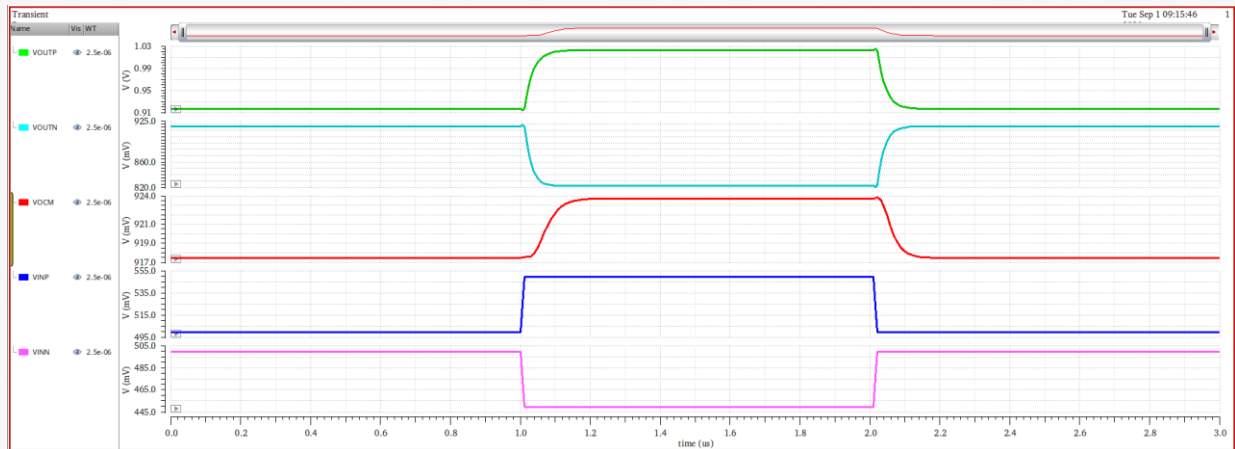
CM LG Simulation results checked

Parameter	Open loop simulation	LG simulation
<i>DC gain</i>	75.96dB	66.41dB
<i>GBW</i>	22.55MHz	7.365MHz

Note that the open loop simulation has much larger gain and GBW.

Closed-Loop OTA Simulation (Transient Analysis)

Differential and CMFB loops stability diff input pulse



■ V_{outP} ■ V_{outN} ■ V_{inP} ■ V_{inN} ■ V_{OCM}

Note that there is no ringing in each of them, and both loops are stable with adequate PM

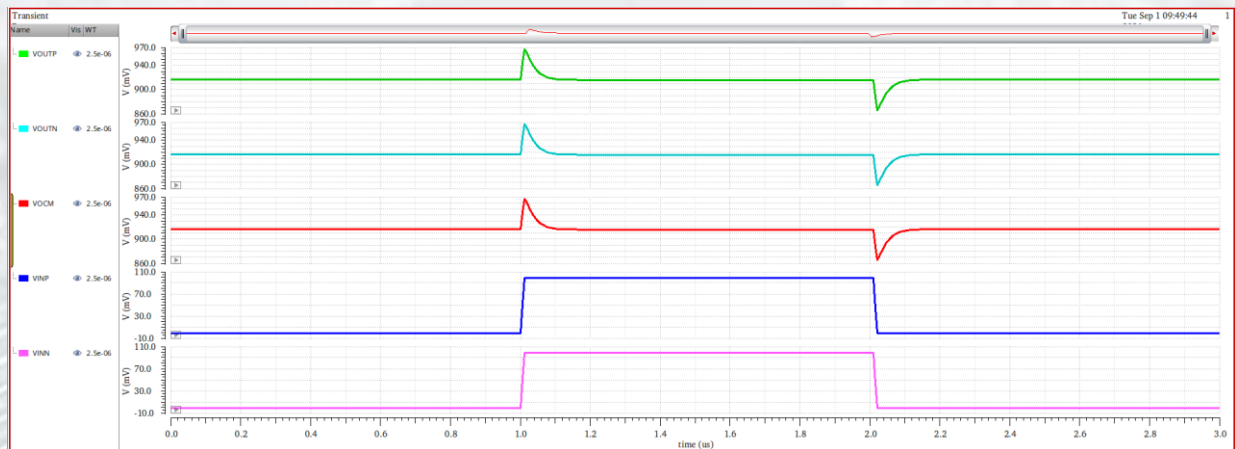
1% settling time

96.45ns

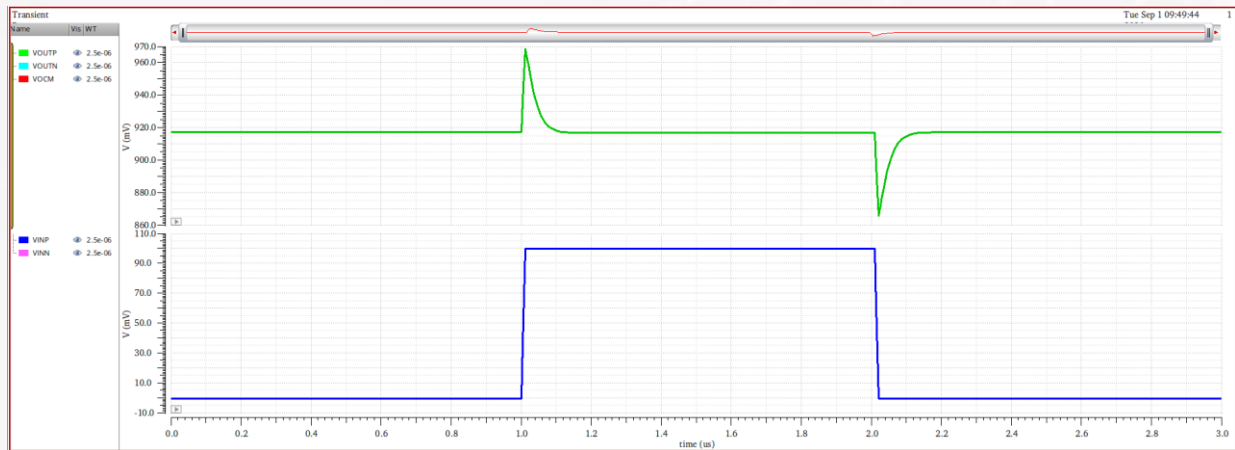
settling time (from 0 → 0.99%) Simulation results

If the specification is not satisfied, we can decrease both C_F and C_S by the same ratio to keep the closed loop gain and increase the settling time.

Differential and CMFB loops stability CM input pulse



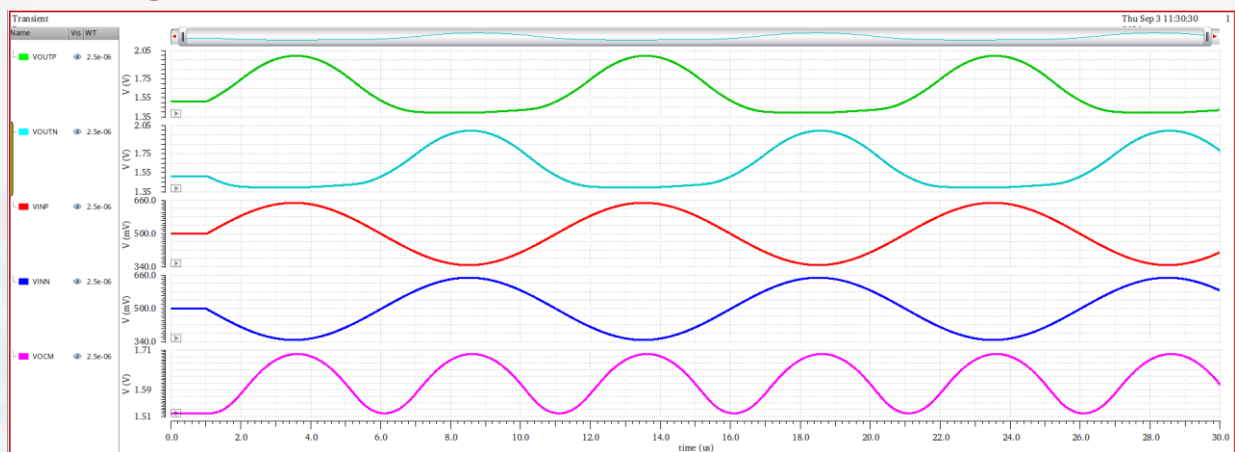
■ V_{outP} ■ V_{outN} ■ V_{inP} ■ V_{inN} ■ V_{OCM}



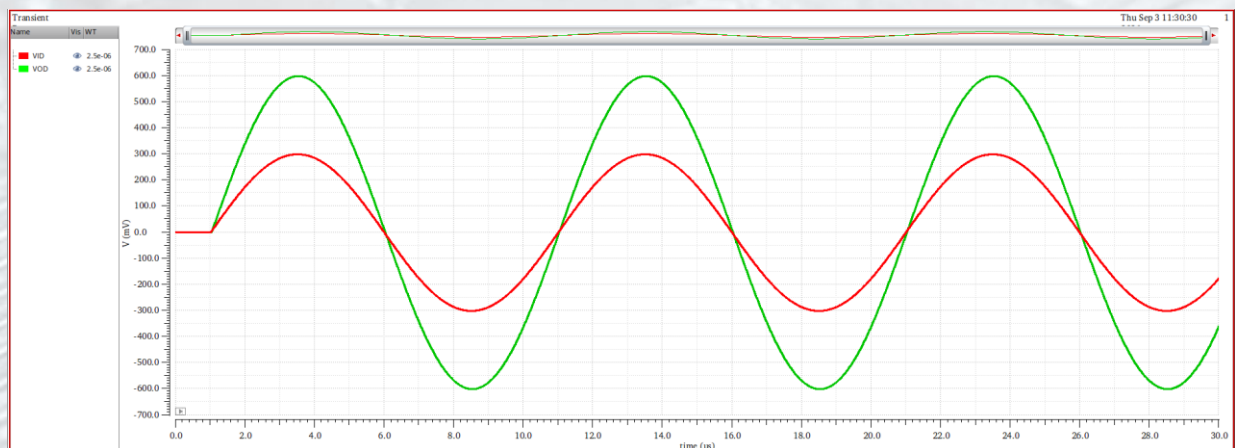
■ $V_{outP}, V_{outN}, V_{outM}$ ■ V_{inP}, V_{inN}

Not that there is no ringing in each of them, and both loops are stable with adequate PM, But the shown response of the output is due to edging effect of the non-ideal input pulse

Output swing

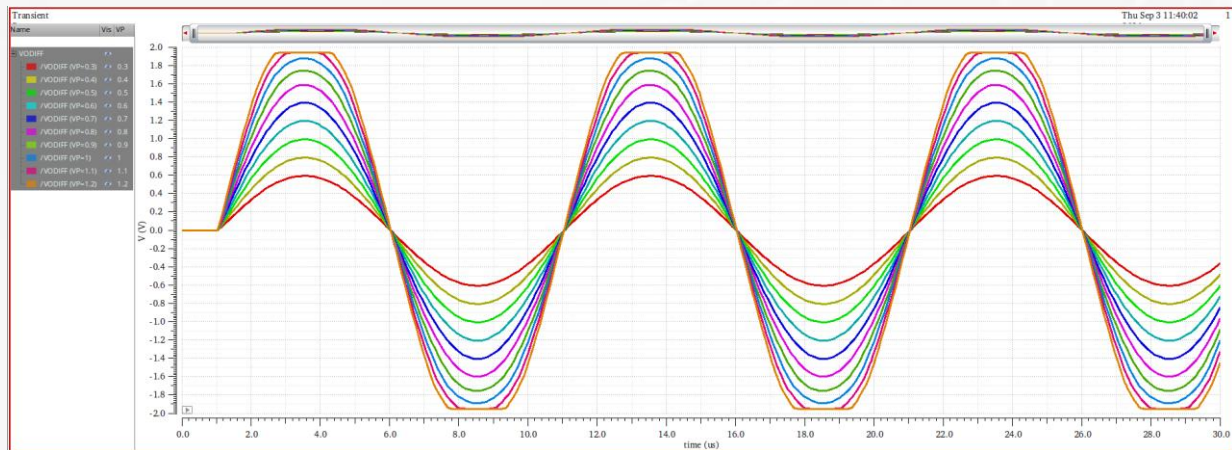


■ V_{outP} ■ V_{outN} ■ V_{inP} ■ V_{inN} ■ V_{outM}



■ V_{od} ■ V_{id}

The results shows that the pk-pk input signal is $0.6V$, and the pk-pk output signal is $1.2V$ that meets the required range.



V_{od} : applying sin inputs with amplitude 300mV:100mV:1.2V

The above graph shows that the maximum pk-pk in the differential output that can be provided is **3.86V** for **2V** pk-pk input signal.